Direction A — WITNESS: nonmalleable endorsement of delegated values

User prompt (trusted)
"Pay the bill in bill.txt"

Agent LLM (unchanged)
reads everything, proposes sink call f(args)

Tool outputs (untrusted)
attacker may APPEND spans

sink call

S1 Intent Slot Compiler
sees ONLY prompt + tool schemas
emits a slot spec per sink arg:
source, type, cardinality, impact

S2 Control Fence
tool and slot must be declared by S1
(data may flow in, control may not)

BLOCK undeclared tool / slot (control influence)

Delegated source D
the observation the slot is delegated to (e.g. bill.txt)

S3 Typed Candidate Enumerator
deterministic: regex + checksum
C = every valid value of the slot type in D
proposal fuzzy-matched -> canonical value

BLOCK proposal not in C (fail-closed)

Trusted state
get_iban, own email, contacts /
user directory, user-initiated txns

S4 Trusted Elimination
drop candidates equal to trusted 'self' values; never reads D's wording

V LLM veto (optional)
quarantined, tool-less
may only ENDORSE -> ESCALATE

seeds

|C| = 1 ?

yes

ENDORSE unique candidate (Prop. 2: sound)

|C| > 1

Counterparty pinning ledger
principal -> value bindings
written ONLY by user choice
(trust-on-first-use)

S5 Trusted Corroboration
exactly one candidate matches a pinned or trusted counterparty?

ENDORSE exactly one corroborated / pinned

none / several

user choice pins value

S6 Closed-Choice Confirmation
defence-rendered, shows ALL candidates
answer = index, never free text
autonomous mode -> ABSTAIN

ESCALATE user picks one, or ABSTAIN

Per-argument label log -> two-sided LIS audit
ENDORSED-unique | ENDORSED-corroborated | ENDORSED-confirmed | BLOCKED-control | BLOCKED-unsupported | ESCALATED | ABSTAINED

trusted input | untrusted input | novel (ours) | adapted from prior work | LLM, veto-only